

- 5.7. For the complementary stage shown in Fig. 5.15, determine the closed-loop gain and noise figure if channel-length modulation is not neglected. Assume matching at the input.

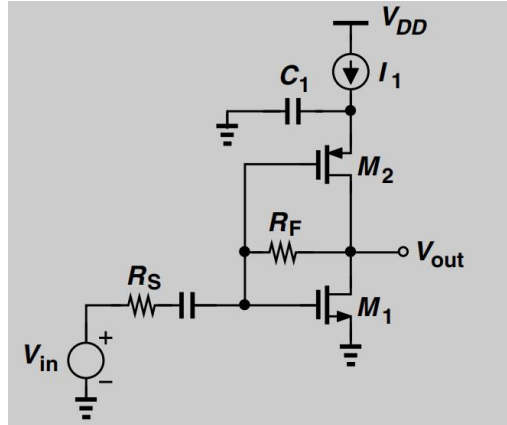


Figure 5.15 CS stage with active load.

- 5.13. The LNA shown in Fig. 5.88 is designed to operate with low supply voltages. Each inductor is chosen to resonate with the total capacitance at its corresponding node at the frequency of interest. Neglect channel-length modulation and body effect and the noise due to the loss in L_2 . Determine the noise figure of the LNA with respect to a source resistance R_S assuming that L_1 can be viewed as a resistance equal to R_p at the resonance frequency. Make sure the result reduces to a familiar form if $R_p \rightarrow \infty$. (Hint: the equivalent transconductance of a degenerated common-source stage is given by $g_m/(1 + g_m R_1)$, where R_1 denotes the degeneration resistance.)

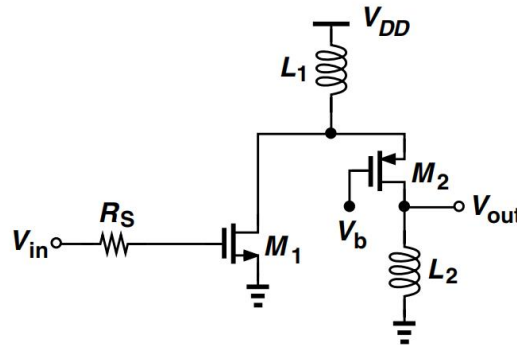


Figure 5.88 Folded-cascode LNA.

- 5.16. Repeat the NF analysis of the CG stage in Fig. 5.43 while including the noise of the feedback network as a voltage, $\overline{V_{nF}^2}$, in series with its input.

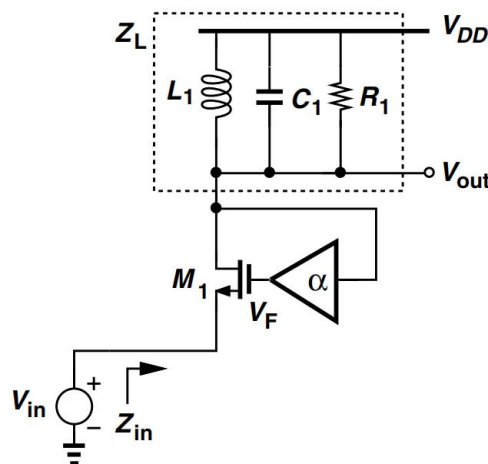


Figure 5.43 CG LNA with feedback.

5.21. Determine the noise figure in Example 5.21 if the gain is reduced by 3 dB.

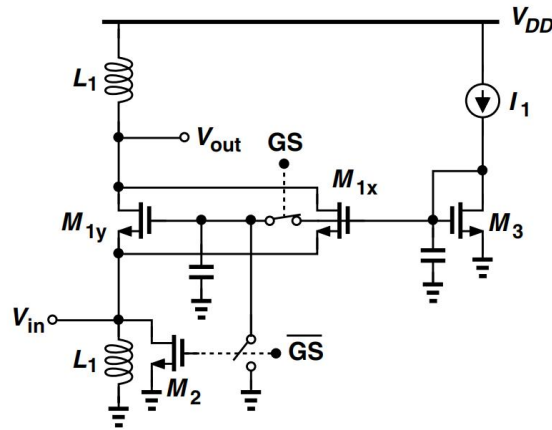


Figure 5.54 Example of gain switching in CG stage.

Example 5.21

Choose the devices in the above circuit for a gain step of 3 dB.

Solution:

To reduce the voltage gain by $\sqrt{2}$, we have

$$\frac{W_{lx}}{W_{lx} + W_{ly}} = \frac{1}{\sqrt{2}}, \quad (5.137)$$

and hence $W_{ly}/W_{lx} = \sqrt{2} - 1$. We also note that, with M_{ly} off, the input resistance rises to $\sqrt{2}R_S$. Thus, $R_{on2} || (\sqrt{2}R_S) = R_S$ and hence

$$R_{on2} = \frac{\sqrt{2}}{\sqrt{2} - 1} R_S. \quad (5.138)$$

In Problem 5.21, we calculate the noise figure after the 3-dB gain reduction.